Review 2 Commitment

UNSIGNED PREPARATION DRAFT. Complete within three working days after Review 2 with both reviewers and supervisor. Counts and M1–M4 ownership are proposed only; names/IDs are intentionally deferred.

1. Project Information

Project code	TBD (PKV working abbreviation)
Project name	PickleVerse – Multiplayer Pickleball Skill-Training and Match Game with AI Coaching
Team	Names and IDs deferred
Supervisor	Lại Đức Hùng (proposal)
Review 2 date	TBD
Reviewer 1	TBD
Reviewer 2	TBD
Version	Draft 0.2 — not approved
Date	23 October 2026

2. Final Registration Form

Final version	Not frozen
Approved at Review 2 (Yes/No)	Pending review
Attached file name	SWP490_PKV_DRAFT_R2_Registration_Source.docx

After this point the project scope does not change.

3. Mandatory Features

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F01	Account and onboarding	Player	Create and verify an account; sign in and load the same server profile; reject invalid credentials; record self-assessment for initial placement.	WF01	M1 TBD
F02	Tutorial and basic controls	Player	Complete or replay rule and movement steps; skipped/completed status persists for that account.	WF01	M1 TBD
F03	Structured skill training	Player	Select an unlocked drill, receive feedback and a 1–3 star result; only the first eligible completion earns the configured reward.	WF01,WF05	M1 TBD
F04	XP levels and cosmetic unlocks	Player	Award XP once per eligible command; recompute level/unlocks and preserve balances after a refresh.	WF05	M1 TBD

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F05	Adaptive AI and skill coaching	Player	Adjust bot challenge and recommend drills from measured skill; validate initial placement and technique classifiers against labelled data.	WF02	M2 TBD
F06	Pickleball movement shots and rules	Player	Serve and execute supported shots while moving; enforce court, kitchen, net, two-bounce and scoring rules in singles and doubles.	WF02,WF03	M2 TBD
F07	Skill queue and ready confirmation	Online player	Match eligible players; show both clients a usable ready dialog; cancel a decliner and requeue consenting players under server policy.	WF03	M3 TBD
F08	Private rooms and teams	Host and invited player	Create/join a coded 1v1 or 2v2 room; validate membership, teams and readiness before start.	WF04	M3 TBD
F09	Online match synchronization	Online players	Synchronize legal movement, shots and score for 1v1/2v2; apply disconnect policy and reject non-member commands.	WF03,WF04	M3 TBD
F10	Durable match result and rating	Player	Finalize an agreed/validated result once; atomically update Elo, wallet, history and rewards; reject conflicting duplicate reports.	WF02,WF03,WF04	M3 TBD
F11	Personal statistics history and replay	Player	Read own progress and recent matches; open authorized replay data and inspect errors and frequently repeated drills.	WF02,WF05	M1 TBD
F12	Friends and global leaderboard	Player	Search and accept/remove friends; read ordered Elo standings and own position; reject unauthorized relationship changes.	WF04	M3 TBD
F13	Daily missions streak and achievements	Player	Advance eligible objectives and streaks; claim each period/reward once and refresh the server-owned state.	WF05	M1 TBD
F14	Offline practice and synchronization	Player	Complete a drill without connectivity and reconcile its permitted progress once after reconnection.	WF05	M1 TBD
F15	Administrator access and user control	Administrator	Authorize Admin only; list/filter users and apply valid ban/restore/role changes with an auditable outcome.	WF06	M4 TBD
F16	Training content management	Administrator	Create/edit and publish a valid lesson/drill configuration which becomes available in the game catalog.	WF06	M4 TBD

ID	Feature	Actor	Acceptance criteria	Workflow	Owner
F17	Match oversight and moderation	Administrator and reporting player	Inspect live/completed matches and disconnects; review reports and appeal decisions with authorized state changes.	WF06	M4 TBD
F18	Operational and learning analytics	Administrator	Show correctly defined active-user, retention, error and queue metrics for an explicit period.	WF06	M4 TBD

4. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
N01	Interactive performance	Median and p95 frame time; input-to-visible feedback; technique feedback delay	Proposed: median ≤16.7 ms and p95 ≤25 ms on an agreed midrange device; input p95 ≤100 ms; coaching <1 s. The proposal states stable 60 FPS; any relaxed percentile needs approval.	Release builds on Windows and Galaxy A32; 15 minute day/night sessions in all four play scenes; capture CPU/GPU/frame-time and touch traces.	M2 TBD
N02	Network responsiveness and capacity	RTT; concurrent players and matches; matchmaking errors	Proposal: typical VN RTT <100 ms; 2,000 players / 500 concurrent matches; no cross-match membership or duplicate result.	Seeded staging multi-client load and network impairment; measure p95 RTT separately from server command latency.	M3 TBD
N03	Availability and recovery	Monthly successful service availability; tested backup/rollback	Proposal: ≥99.5% monthly availability; proposed recovery targets RPO ≤24 h / RTO ≤4 h require approval.	External probes plus service metrics; restore a staging backup and deploy/rollback a known version.	M4 TBD

ID	Quality attribute	Metric	Threshold	Measurement method	Owner
N04	Platform and usable controls	Acceptance pass matrix; safe-area clipping; target hit area	Windows 10+, Android 9+, iOS 14+ as proposed; all main flows pass; no clipped essential controls; proposed touch targets ≥48 dp.	Physical Windows/Android/iOS test matrix and landscape safe-area screenshots; simulation does not prove physical performance.	M2 TBD
N05	Security and trusted data	Unauthorized actions; replayed commands; transport/at-rest protection; simulation authority	JWT access control; reject all unauthorized mutation/replayed conflicting commands; proposal AES-256 and dedicated authority require deployment/design evidence.	Auth/RBAC negative tests, duplicate/race tests, TLS/secret/backup configuration inspection and malicious-peer simulation.	M4 TBD
N06	Offline data and learning quality	Offline completion/sync duplication; classifier and recommendation quality	Proposal offline training plus exactly-once permitted sync; proposed ML macro-F1 ≥0.80 and recommendation acceptance ≥80% require labelled evaluation and agreement.	Disconnect/reconnect drill tests; account-switch conflicts; held-out labelled dataset with confusion matrix and recommendation review.	M1 TBD

5. Out of Scope

Item	Reason
Tournaments/spectators/ranked ladders	Proposal exclusions; clarify boundary with included Elo ratings
Video swing analysis	Proposal exclusion
Real-money payment	Post-capstone extension

6. Workflow Commitments

Minimum number of main workflows = number of team members. Each workflow has exactly one owner. Each mandatory feature belongs to at least one workflow.

6.1 Workflow WF01

Workflow name	Register learn and complete a first drill
Owner (one member)	M1 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Player, SMTP delivery service
Trigger	Submit and verify account
Business outcome	A verified player completes a first drill and sees the persisted result.
Main entity state path	Account registered → verified → tutorial completed → training attempt completed
Activity diagram (Report 4 section)	SDD section 6; ACT-WF01

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F01	Account and onboarding	Create and verify an account; sign in and load the same server profile; reject invalid credentials; record self-assessment for initial placement.	TC-F01-01..03
F02	Tutorial and basic controls	Complete or replay rule and movement steps; skipped/completed status persists for that account.	TC-F02-01..03
F03	Structured skill training	Select an unlocked drill, receive feedback and a 1–3 star result; only the first eligible completion earns the configured reward.	TC-F03-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF01-MAIN	Main	Seeded staging	Submit and verify account; Open tutorial and perform controls; Choose an unlocked drill; Start server attempt and submit measurements; Complete and refresh stars and first reward	A verified player completes a first drill and sees the persisted result.	M1
WF01-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	Expired OTP or invalid trial data is rejected; account/attempt can be retried without duplicate reward.	Reject and preserve consistent state	M1

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	9	6 proposed; 0 verified	M1
E2E workflow	1	0 proposed; 0 verified	M1
State transitions	29	13 proposed; 0 verified	M1
NFR acceptance	0	0 proposed; 0 verified	M1
Total	39	19 proposed; 0 verified	M1

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

6.2 Workflow WF02

Workflow name	Practice against an adaptive bot
Owner (one member)	M2 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Player
Trigger	Choose bot difficulty and court
Business outcome	The player receives a legal match result and coaching evidence.
Main entity state path	Match Intro → PreServe → Rally → MatchOver → result persisted
Activity diagram (Report 4 section)	SDD section 6; ACT-WF02

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F05	Adaptive AI and skill coaching	Adjust bot challenge and recommend drills from measured skill; validate initial placement and technique classifiers against labelled data.	TC-F05-01..03

Feature ID	Feature	Acceptance criteria	Test case IDs
F06	Pickleball movement shots and rules	Serve and execute supported shots while moving; enforce court, kitchen, net, two-bounce and scoring rules in singles and doubles.	TC-F06-01..03
F10	Durable match result and rating	Finalize an agreed/validated result once; atomically update Elo, wallet, history and rewards; reject conflicting duplicate reports.	TC-F10-01..03
F11	Personal statistics history and replay	Read own progress and recent matches; open authorized replay data and inspect errors and frequently repeated drills.	TC-F11-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF02-MAIN	Main	Seeded staging	Choose bot difficulty and court; Run entry cinematic; Play serve and rally with legal inputs; Resolve points and match end; Wait for end notice then show player result and statistics	The player receives a legal match result and coaching evidence.	M2
WF02-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	Illegal kitchen/serve action is penalized; an aborted/unverified result cannot award a duplicate reward.	Reject and preserve consistent state	M2

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	6	0 proposed; 0 verified	M2
E2E workflow	1	0 proposed; 0 verified	M2
State transitions	14	14 proposed; 0 verified	M2
NFR acceptance	2	0 proposed; 0 verified	M2
Total	23	14 proposed; 0 verified	M2

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN

Criterion	Committed
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

6.3 Workflow WF03

Workflow name	Find accept and finish an online match
Owner (one member)	M3 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Online player A, Online player B, Photon service
Trigger	Join eligible skill queue
Business outcome	A canonical match result updates each participant once.
Main entity state path	Queue searching → offer → all accepted → playing → reported → finished
Activity diagram (Report 4 section)	SDD section 6; ACT-WF03

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F06	Pickleball movement shots and rules	Serve and execute supported shots while moving; enforce court, kitchen, net, two-bounce and scoring rules in singles and doubles.	TC-F06-01..03
F07	Skill queue and ready confirmation	Match eligible players; show both clients a usable ready dialog; cancel a decliner and requeue consenting players under server policy.	TC-F07-01..03
F09	Online match synchronization	Synchronize legal movement, shots and score for 1v1/2v2; apply disconnect policy and reject non-member commands.	TC-F09-01..03
F10	Durable match result and rating	Finalize an agreed/validated result once; atomically update Elo, wallet, history and rewards; reject conflicting duplicate reports.	TC-F10-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF03-MAIN	Main	Seeded staging	Join eligible skill queue; Poll canonical offer and accept; Wait for all required human acceptances; Start Photon Shared session and play; Report consistent results and read finalized rating	A canonical match result updates each participant once.	M3

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF03-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	A decline/timeout cancels the offer; a conflicting report blocks normal finalization and applies server policy.	Reject and preserve consistent state	M3

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	9	6 proposed; 0 verified	M3
E2E workflow	1	0 proposed; 0 verified	M3
State transitions	34	20 proposed; 0 verified	M3
NFR acceptance	1	0 proposed; 0 verified	M3
Total	45	26 proposed; 0 verified	M3

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

6.4 Workflow WF04

Workflow name	Create a private team match
Owner (one member)	M3 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Room host, Invited player
Trigger	Find a friend or share room code
Business outcome	A coded room starts with the declared 1v1 or 2v2 roster.
Main entity state path	Room created → joined → teams ready → playing → finished

Activity diagram (Report 4 section)	SDD section 6; ACT-WF04

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F08	Private rooms and teams	Create/join a coded 1v1 or 2v2 room; validate membership, teams and readiness before start.	TC-F08-01..03
F09	Online match synchronization	Synchronize legal movement, shots and score for 1v1/2v2; apply disconnect policy and reject non-member commands.	TC-F09-01..03
F10	Durable match result and rating	Finalize an agreed/validated result once; atomically update Elo, wallet, history and rewards; reject conflicting duplicate reports.	TC-F10-01..03
F12	Friends and global leaderboard	Search and accept/remove friends; read ordered Elo standings and own position; reject unauthorized relationship changes.	TC-F12-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF04-MAIN	Main	Seeded staging	Find a friend or share room code; Create selected room mode; Join and assign valid teams; Confirm ready roster and start; Finish and inspect authorized statistics	A coded room starts with the declared 1v1 or 2v2 roster.	M3
WF04-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	Wrong/full room or unauthorized team/start command is rejected; the room stays consistent.	Reject and preserve consistent state	M3

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	6	3 proposed; 0 verified	M3
E2E workflow	1	0 proposed; 0 verified	M3
State transitions	13	6 proposed; 0 verified	M3
NFR acceptance	0	0 proposed; 0 verified	M3
Total	20	9 proposed; 0 verified	M3

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN

Criterion	Committed
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

6.5 Workflow WF05

Workflow name	Review progress and claim rewards
Owner (one member)	M1 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Player
Trigger	Refresh server progress
Business outcome	The player sees next practice, history and exactly-once rewards.
Main entity state path	Attempt completed → progress refreshed → reward eligible → claimed
Activity diagram (Report 4 section)	SDD section 6; ACT-WF05

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F03	Structured skill training	Select an unlocked drill, receive feedback and a 1–3 star result; only the first eligible completion earns the configured reward.	TC-F03-01..03
F04	XP levels and cosmetic unlocks	Award XP once per eligible command; recompute level/unlocks and preserve balances after a refresh.	TC-F04-01..03
F11	Personal statistics history and replay	Read own progress and recent matches; open authorized replay data and inspect errors and frequently repeated drills.	TC-F11-01..03
F13	Daily missions streak and achievements	Advance eligible objectives and streaks; claim each period/reward once and refresh the server-owned state.	TC-F13-01..03
F14	Offline practice and synchronization	Complete a drill without connectivity and reconcile its permitted progress once after reconnection.	TC-F14-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF05-MAIN	Main	Seeded staging	Refresh server progress; Inspect stars and error aggregates; Open suggested drill or replay; Claim eligible mission/achievement reward; Refresh wallet and continuation target	The player sees next practice, history and exactly-once rewards.	M1
WF05-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	A repeated claim does not credit again; offline practice/sync is a required gap and must be demonstrated after implementation.	Reject and preserve consistent state	M1

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	12	6 proposed; 0 verified	M1
E2E workflow	1	0 proposed; 0 verified	M1
State transitions	10	10 proposed; 0 verified	M1
NFR acceptance	1	0 proposed; 0 verified	M1
Total	24	16 proposed; 0 verified	M1

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

6.6 Workflow WF06

Workflow name	Administer users content and operations
Owner (one member)	M4 — named assignment pending
Supporting members	TBD after team confirmation
Actors	Administrator, Reporting player
Trigger	Authenticate as administrator
Business outcome	Authorized actions change service data and support operational decisions.
Main entity state path	User active → sanctioned/restored; lesson draft → published; report pending → resolved
Activity diagram (Report 4 section)	SDD section 6; ACT-WF06

Features in this workflow

Feature ID	Feature	Acceptance criteria	Test case IDs
F15	Administrator access and user control	Authorize Admin only; list/filter users and apply valid ban/restore/role changes with an auditable outcome.	TC-F15-01..03
F16	Training content management	Create/edit and publish a valid lesson/drill configuration which becomes available in the game catalog.	TC-F16-01..03
F17	Match oversight and moderation	Inspect live/completed matches and disconnects; review reports and appeal decisions with authorized state changes.	TC-F17-01..03
F18	Operational and learning analytics	Show correctly defined active-user, retention, error and queue metrics for an explicit period.	TC-F18-01..03

Demo scenarios

Scenario ID	Type	Preconditions and data	Main steps	Expected result	Presenter
WF06-MAIN	Main	Seeded staging	Authenticate as administrator; Inspect a user or moderation report; Apply valid access/status decision; Create and publish playable lesson configuration; Inspect period-defined operational analytics	Authorized actions change service data and support operational decisions.	M4
WF06-EXCEPTION	Exception	Seeded invalid/timeout/foreign account	A non-admin request is denied; invalid publication/status data leaves existing records unchanged.	Reject and preserve consistent state	M4

Committed test cases

Test type	Count	Automated in CI	Responsible member(s)
Feature tests	12	3 proposed; 0 verified	M4
E2E workflow	1	0 proposed; 0 verified	M4
State transitions	18	5 proposed; 0 verified	M4
NFR acceptance	2	0 proposed; 0 verified	M4
Total	33	8 proposed; 0 verified	M4

Pass criteria at Defense 1.1

Criterion	Committed
End-to-end staging	Target yes; NOT RUN
Main scenario	Target pass; NOT RUN
Exception scenarios	Target ≥80% overall; NOT RUN
Workflow tests	Target ≥90%; 100% mandatory main path
Open Critical/High	Target 0; current findings unresolved

Reviewer note

Note	Pending reviewers; no conclusion entered.

7. Individual Commitments

One block per member. Every member owns at least one main workflow and 2–3 parts. Each test case and each fix is counted for one member only.

7.1 Proposed role slot M1

Full name	Deferred by user
Student ID	Deferred by user
Role in team	Identity training and progression (proposed)

Ownership

Type	Items (IDs / names)	Notes
Main workflows	WF01, WF05	Named confirmation pending
Mandatory features	F01, F02, F03, F04, F11, F13, F14	Primary responsibility proposed
Modules/entities	Account and tutorials; Training and player progression	Two parts; confirm ownership
Design decisions	D1, D4	Proposed
NFRs	N06	Proposed

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Unique test cases	63	TBD
Automated in CI	35 proposed; 0 verified	TBD
Demo scenarios	4	TBD
Required fixes	See section 9; names not assigned	TBD
Documentation	SRS/SDD sections for owned workflows	TBD

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
Not reviewed	To be completed by reviewers	

Member confirmation

Signature / date	Unsigned — member confirmation pending

7.2 Proposed role slot M2

Full name	Deferred by user
Student ID	Deferred by user
Role in team	Local gameplay and AI (proposed)

Ownership

Type	Items (IDs / names)	Notes
Main workflows	WF02	Named confirmation pending
Mandatory features	F05, F06	Primary responsibility proposed
Modules/entities	Rules and match presentation; Bot coaching and performance	Two parts; confirm ownership
Design decisions		Proposed
NFRs	N01, N04	Proposed

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Unique test cases	23	TBD
Automated in CI	14 proposed; 0 verified	TBD
Demo scenarios	2	TBD
Required fixes	See section 9; names not assigned	TBD
Documentation	SRS/SDD sections for owned workflows	TBD

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
Not reviewed	To be completed by reviewers	

Member confirmation

Signature / date	Unsigned — member confirmation pending

7.3 Proposed role slot M3

Full name	Deferred by user
Student ID	Deferred by user
Role in team	Online play and social (proposed)

Ownership

Type	Items (IDs / names)	Notes
Main workflows	WF03, WF04	Named confirmation pending
Mandatory features	F07, F08, F09, F10, F12	Primary responsibility proposed
Modules/entities	Sessions and rooms; Results and social integration	Two parts; confirm ownership
Design decisions	D3	Proposed
NFRs	N02	Proposed

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Unique test cases	65	TBD
Automated in CI	35 proposed; 0 verified	TBD
Demo scenarios	4	TBD
Required fixes	See section 9; names not assigned	TBD
Documentation	SRS/SDD sections for owned workflows	TBD

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
Not reviewed	To be completed by reviewers	

Member confirmation

Signature / date	Unsigned — member confirmation pending

7.4 Proposed role slot M4

Full name	Deferred by user
Student ID	Deferred by user

Role in team	Backend administration and operations (proposed)

Ownership

Type	Items (IDs / names)	Notes
Main workflows	WF06	Named confirmation pending
Mandatory features	F15, F16, F17, F18	Primary responsibility proposed
Modules/entities	Admin and catalog services; Deployment security and observability	Two parts; confirm ownership
Design decisions	D2, D5	Proposed
NFRs	N03, N05	Proposed

Delivery commitments to Defense 1.1

Commitment	Committed (count / IDs)	Due
Unique test cases	33	TBD
Automated in CI	8 proposed; 0 verified	TBD
Demo scenarios	2	TBD
Required fixes	See section 9; names not assigned	TBD
Documentation	SRS/SDD sections for owned workflows	TBD

Ownership verification at Review 2 (completed by reviewers)

Result	Weak points observed	Follow-up (by whom, by when)
Not reviewed	To be completed by reviewers	

Member confirmation

Signature / date	Unsigned — member confirmation pending

8. Summary and Cross-check

8.1 By workflow

Workflow	Owner	Features	Demo scenarios (main / exception)	Test cases	Automated
WF01	M1	F01,F02,F03	1 / 1	39	19 proposed
WF02	M2	F05,F06,F10,F11	1 / 1	23	14 proposed
WF03	M3	F06,F07,F09,F10	1 / 1	45	26 proposed
WF04	M3	F08,F09,F10,F12	1 / 1	20	9 proposed
WF05	M1	F03,F04,F11,F13,F14	1 / 1	24	16 proposed
WF06	M4	F15,F16,F17,F18	1 / 1	33	8 proposed
Unique total		18	6 / 6	184	92 proposed; 0 verified

8.2 By member

Member	Workflows owned	Features	Test cases	Automated	Demo scenarios	Review 2 verification
M1	WF01,WF05	7	63	35 proposed	4	Not reviewed
M2	WF02	2	23	14 proposed	2	Not reviewed
M3	WF03,WF04	5	65	35 proposed	4	Not reviewed
M4	WF06	4	33	8 proposed	2	Not reviewed
Unique total	6	18	184	92 proposed	12	Pending

Workflow and member test totals both equal 184 unique cases; proposed CI totals both equal 92, with zero executions verified. Feature totals count unique IDs; shared workflow coverage is not added twice.

8.3 Committed Numbers

Number of main workflows	6 proposed
Number of demo scenarios (main / exception)	6 / 6 proposed
Committed functional test cases	184 proposed; unsigned
Of which automated in CI	At least 92 proposed; no run verified

8.4 Eligibility Thresholds for the Viva

Criterion	Minimum threshold	Agreed threshold
Main workflows	*100% run end-to-end*	Pending agreement; retain minimum
Demo scenarios	*100% main; ≥ 80% exception*	Pending agreement; retain minimum
Test case count	*≥ committed number*	Pending agreement; retain minimum
Test pass rate	*≥ 90% overall; 100% main-path tests of mandatory features*	Pending agreement; retain minimum
Automation	*≥ 50% run in CI*	Pending agreement; retain minimum
Defects	*No open Critical or High defects*	Pending agreement; retain minimum
Individual	*Each member demos the workflow they own and meets their delivery commitments*	Pending agreement; retain minimum

9. Required Fixes

Items from the Review 2 minutes and Planned items from Form BM02.

ID	Item	Related workflow	Owner	Due
SR01	Photon Fusion Shared gives a peer state authority. This is not the proposed trusted dedicated simulation and leaves host/peer trust risk.	Registration vs SDD actual design	Named owner TBD	TBD
SR02	The active progression runtime requires server state. Offline drill execution and exactly-once later synchronization are not demonstrated.	F05 F14 and N06	Named owner TBD	TBD
SR03	Initial-placement ML, telemetry technique classifier, labelled evaluation and validated recommendations are not evidenced. Rule-based feedback does not establish those commitments.	F05 and N06	Named owner TBD	TBD
SR04	The React login sends emailOrUsername/passwordHash and reads role/token, while the API requires Email/Password and returns AccessToken/Profile. The default contracts are incompatible.	F15 administrator sign in	Named owner TBD	TBD
SR05	Admin mutations serialize enum status as Banned/Active while the SQL constraint uses ACTIVE/BANNED/SUSPENDED; these values can fail the database constraint.	F15 user status mutation	Named owner TBD	TBD

ID	Item	Related workflow	Owner	Due
SR06	Admin lesson creation omits catalog Code/Config, while the game catalog selects coded configured lessons. CRUD does not yet demonstrate publishing a playable drill.	F16 content publication	Named owner TBD	TBD
SR07	Complete live-match oversight, appeals and operational analytics are not evidenced. ActiveUsersToday currently counts all users rather than dated activity.	F17 F18 administration	Named owner TBD	TBD
SR08	Current-revision physical-device, network/load, security, availability, offline and ML acceptance results are unavailable. Test source is not execution evidence.	N01 through N06 and Report 5	Named owner TBD	TBD
SR09	The local review-2 tag and CI execution evidence are absent; the required official tagged self-review and automated-in-CI target cannot be verified.	Review 2 baseline and CI	Named owner TBD	TBD
SR10	The approved registration revision, official project code and decisions accepting stack/authority deviations are unconfirmed. No scope reduction has been assumed.	BM03 and registration baseline	Named owner TBD	TBD
SR11	Server-owned transactions and matching human reports improve result integrity, but peers still supply simulation outcomes and training measurements. Agreement does not independently prove legal physics.	Online result validation	Named owner TBD	TBD
SR12	Real friendship APIs and room-code play exist and cover the proposal room-code requirement. Account-targeted invitations are an unverified desired UX extension; confirm whether they should become a separate commitment.	Optional account invitation UX	Named owner TBD	TBD
SR13	Fallback authentication configuration, deployed TLS/AES-256 controls, production rollback and iOS build acceptance require deployment evidence; configuration values are intentionally redacted.	Deployment secrets and platforms	Named owner TBD	TBD

10. Review 2 Conclusion (completed by reviewers)

Reviewers to complete: Pass / Conditional pass / Re-review. No option selected.

11. Confirmation

Role	Full name	Signature	Date
Reviewer 1			
Reviewer 2			

Role	Full name	Signature	Date
Supervisor	Lại Đức Hùng (proposal)		
Team leader			